FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. D

Sol. $\frac{1}{V_1}-\frac{2}{-15}=\frac{1-2}{-10}\Rightarrow V_1=-30\,\text{cm}$ and $h=5\left(\frac{8/5}{2}\right)=4\,\text{cm}$

So seperation between two images $=14\text{cm}$

2. A

Sol. $\frac{\lambda}{2\pi\varepsilon_0 x}=\frac{2q\ell}{2\pi\varepsilon_0 x^3}\Rightarrow x^2=\frac{2q\ell}{\lambda}\Rightarrow x=2$

3. C

Sol. Buoyant force will become $=V\rho\left(g+a\right)$ and the sudden increment in it will provide the required acceleration to block.

4. D

Sol. M.I. about diagonal of a square is $\frac{ma^2}{12}$.

5. AC

Sol. $F-umg=ma\Rightarrow F=2N$

$T_F=2\times\frac{1}{2}=1\,\text{N-m}$

$$\vec{T} = \vec{T}_F + \vec{T}_g + \vec{T}_f + \vec{T}_N = 1 + 10 \times \frac{5}{2} - 10 \times \frac{51}{20} = \frac{1}{2} \text{ N-m}$$

6. AD

Sol. As the point of application of friction does not move so work done by friction force is zero.

7. BC

Sol. Potential of the centre of sphere $= \frac{kq}{r}$

When earthed $V = 0 = \frac{kq}{r} + \frac{k(q')}{R} \Rightarrow q' = \frac{R}{r}(-q)$

8. AD

Sol. Using work energy theorem

$$-\mu mgx - \frac{1}{2}kx^2 = -\frac{1}{2}mv^2$$

$$11x + 25x^2 - 36 = 0$$

$$\Rightarrow x = 1m$$

Now when it stops momentary for the second time

$$-\mu mgd - \left[\frac{1}{2}k\,d_2^2 - \frac{1}{2}k\,d_1^2\right] = 0$$

$$11d + 25d_2^2 - 25d_1^2 = 0$$

$$11d_1 + 11d_2 + 25d_2^2 - 25d_1^2 = 0$$

So, $25d_2^2 + 11d_2 - 14 = 0 \left[as\, d = 1m\right]$

$$d_2 = 0.56m$$

So acceleration $a = \frac{kd_2 - \mu mg}{m}$

$$a = 8.5\, m/s^2$$

9. AC

10. ABC

Sol. $x_{CM} = \frac{2 \times 15}{5} = 6$

$$y_{CM} = \frac{3 \times 10}{5} = 12$$

11. BC

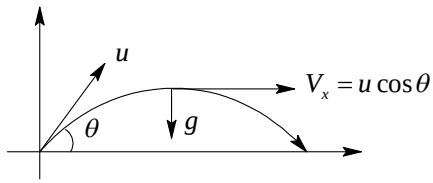
12. BCD

Sol. At highest point

$V_y = 0$

$V_x = u\cos\theta$

Hence $\overline{V} \perp \overline{a}$ and speed is maximum.



SECTION – C

13. 00001.57

Sol. $K_{eq}=\dfrac{16K}{5}$

$K_{eq}=\dfrac{16K}{5}$

Use $T=2\pi\sqrt{\dfrac{m}{K_{eq}}}$

14. 00019.20

Sol. $v+u=120-----(1)$

$\dfrac{v^2}{u^2}=\dfrac{1}{16}\Rightarrow\dfrac{v}{u}=\dfrac{1}{4}----(2)$

$\Rightarrow v=24$ and $u=96$

$\Rightarrow\dfrac{1}{f}=\dfrac{1}{v}-\dfrac{1}{u}=\dfrac{1}{24}+\dfrac{1}{96}$

$\Rightarrow f=19.2\,cm$

15. 00001.80

Sol. $F=\dfrac{9\times10^9\times2\times10^{-6}\times10^{-6}}{\left(10\times10^{-2}\right)^2}=1.8\,\text{Newton}$

16. 00006.28

Sol. $Y=\dfrac{mg/A}{\Delta\ell/\ell}$ at equilibrium

$\Rightarrow T-mg=ma$ at general position where $Y=\dfrac{T/A}{\left(\Delta\ell+x\right)/\ell}$

$\Rightarrow\dfrac{YA\Delta\ell}{\ell}+\dfrac{YAx}{\ell}-mg=ma$

$\Rightarrow a=\dfrac{YA}{m\ell}(-x)$ so time period $=2\pi\sqrt{\dfrac{m\ell}{YA}}=6.28\,\text{sec}$

17. 00019.94

Sol. Process 1 – 2 and 3 – 4 are isochoric so no work in these processes. Other two processes are isobaric

Nett work done $=\Delta W_{2-3}+\Delta W_{4-1}$

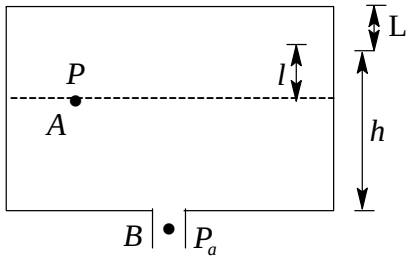
$=nR\left(T_3-T_2\right)+nR\left(T_1-T_4\right)$

$=nR\left[T_3-T_2+T_1-T_4\right]$

$=3\times8.31\left[800\right]=19.94KJ$

18. 00075.00

Sol. $P_a + 0 + 0 = P + \rho g\left(h - l\right) + 0$



Here l is the distance by which the water level has descended. $\left(l = 0.5\,m\right)$

$P_a\left(AL\right) = PA\left(L + l\right)$

As process can be taken as isothermal for air,

$P_a\left(AL\right) = PA\left(L + l\right)$

$L\left(P_a - P\right) = Pl$

$L = \dfrac{\left[P_a - \rho gh + \rho gl\right]l}{\rho g\left(h - l\right)}$

$= \dfrac{10^4\left[10 - 4.5 + 0.5\right]l}{10^4\left[4.5 - 0.5\right]}$

$= 0.75\,cm$

$= 75\,cm$

Chemistry

PART – II

SECTION – A

19. A

20. C

21. B

22. C

Sol. $r = k[A][B_2]$ & $k_c = \dfrac{[A][A]}{[A_2]}$

$\therefore [A] = K^{1/2}[A_2]^{1/2}$

Thus $r = K.K_c^{1/2}[A_2]^{1/2}[B_2] = K'[A_2]^{1/2}[B_2]$

Hence order reaction s 3/2

23. ABCD

24. CD

25. ABC

26. ABC

27. ABCD

28. ABD

29. ABC

30. ABCD

SECTION – C

31. 00009.11

Sol. $Q_{rev.} = nRT\ln\left(\dfrac{V_2}{V_1}\right) = 1\times 8.3\times 298\times 2.303\times 0.477 = 2717.67\ J$

$Q_{rev.} = 2717.67$

$\Delta S_{system} = \dfrac{Q_{rev.}}{T} = 9.11$

32. 00345.00

Sol. Balance eq. is

$4Au + 8CN^- + O_2 + 2H_2O \rightarrow 4Au(CN)_2^- + 4OH^-$

(i) $4Au \rightarrow 4Au^+ + 4e$ $\quad \Delta G^\circ = 4\times 96500\times 1.5\ J$

$4Au^+ + 8CN^- \rightarrow 4Au(CN)_2^-$ $\quad \Delta G^\circ = -2\times 8\times 300\times 38\times 4\ J$

$4Au + 8CN^- \rightleftharpoons 4Au(CN)_2^- + 4e$ $\quad \Delta G_J^\circ = 579 - 729.6\ KJ = -150.6\ KJ$

(ii) $O_2 + 4H^+ + 4e \rightarrow 2H_2O$ $\quad \Delta G'^\circ = -4\times 96500\times 1.2\ J$

$4H_2O \rightarrow 4H^+ + 4OH^-$ $\quad \Delta G_2'^\circ = 2\times 8\times 300\times 56$

$O_2 + 2H_2O + 4e \rightarrow 4OH^-$ $\quad \Delta G_3'^\circ = -463.2 + 268.8\ kJ = -194.4\ kJ$

$\Delta G^\circ_{reaction} = \Delta G_3^\circ + \Delta G_3'^\circ = -345\ kJ$

33. 00000.01

Sol. $HCOOH \rightleftharpoons HCOO^- + H^+$

$$K_{a_1} = \frac{\left[HCOO^-\right]\left[H^+\right]}{\left[HCOOH\right]} = \frac{\left[H^+\right]^2}{\left[HCOOH\right]}$$

$CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

$$K_{a_2} = \frac{\left[CH_3COO^-\right]\left[H^+\right]}{\left[CH_3COOH\right]} = \frac{\left[H^+\right]^2}{\left[CH_3COOH\right]}$$

As $\left[H^+\right]$ in both case are equal

$K_{a_1} \times [HCOOH] = K_{a_2}[CH_3COOH]$

$$\therefore \quad [CH_3COOH] = \frac{K_{a_1}}{K_{a_2}} \times [HCOOH] = \frac{1.8\times10^{-4}}{1.8\times10^{-5}} \times 10^{-3} = 0.01\,M$$

34. 00871.10

Sol. $p_{O_2} + p_{H_2O} = 740$

$p_{O_2} = 740 - 31.8$ at $30^\circ C$

= 708.2 torr

$p_{N_2} = 780 - 23.8$ at $25^\circ C$

= 756.2 torr

Mix at 35^oC

$$\frac{p_1V_1}{T_1} + \frac{p_2V_2}{T_2} = \frac{pV}{T}$$

$$\frac{708.2\times250}{303} + \frac{756.2\times300}{298} = \frac{p\times500}{308}$$

p = 828.9 torr

Final pressure = p + aqueous tension = 828.9 + 42.2 = 871.1

35. 00020.48

Sol. $$\lambda_M = \frac{K\times1000}{M} \Rightarrow (80\times2+40) = \frac{\left(2.00\times10^{-6} - 0.40\times10^{-6}\right)\times1000}{M}$$

So, $M = 8\times10^{-6}$

$K_{sp} = 2048\times10^{-18} = 20.48\times10^{-16}$

36. 00013.68

Sol. $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$

14 38.5

0 10.5 14

So 25-24.5=0.5 ml of H_2O vap is present

So, vap. Pressure of water at this temperature= $\frac{0.5}{25} = \frac{1}{50}$ atm

Now apply $\frac{P_A^o - P_{solution}}{P_A^o} = 0.1$ and $P_A^o = \frac{1}{50}$

$\Rightarrow P_{solution} = 13.68 mm\ of\ Hg$

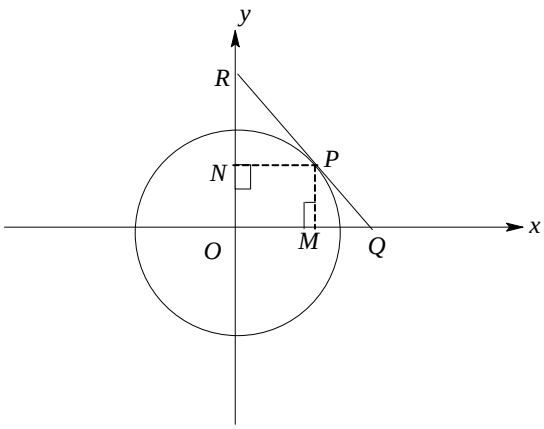
Mathematics

PART – III

SECTION – A

37. D

Sol.



Consider a unit circle having centre at origin.

$ON = PM = \sin\theta \qquad PQ = \tan\theta \qquad OQ = \sec\theta \qquad OP = 1$

$OM = PN = \cos\theta \qquad PR = \cot\theta \qquad OR = \cos ec\,\theta \qquad MN = 1$

$\therefore f(\theta) = PM + PN + PQ + PR - OQ - OR$

$= PM + PN + QR - OQ - OR$

$= PM + PN + QR - (OM + MQ) - (ON + NR)$

$= QR - MQ - NR$

$< MN$

(Since, $NR + MN + MQ > QR$)

$\therefore f(\theta) < 1$

38. D

Sol. $P(x) = (x-(1-i))(x-(1+i))(x-(2-2i))(x-(2+2i))(x^2 - cx + 4)$

Since the first four roots are all distinct, the term $(x^2 - cx + 4)$ must be a product of any combination of two (not necessarily distinct) factors from the set $(x-(1-i)), (x-(1+i))$ $(x-(2-2i))$ $(x-(2+2i))$. The only combinations are

$(x-(1-i))\&(x-(2+2i))$ *or* $(x-(1+i))\&(x-(2-2i))$

$\therefore c = 3 \pm i$

$|c| = \sqrt{10}$

39. A

Sol. $\alpha + \beta + \gamma = 0, \frac{\alpha}{2} + \beta + 2\gamma = 0$

$$\Rightarrow \frac{\alpha}{2}=\gamma \Rightarrow \beta+3\gamma=0 \Rightarrow \gamma=-\frac{\beta}{3}$$

$$\frac{\alpha}{2}=\gamma=-\frac{\beta}{3}$$

$$\alpha=2\gamma, \beta=-3\gamma$$

$$p=\alpha\beta+\beta\gamma+\gamma\alpha=-6\gamma^2+3\gamma^2+2\gamma^2=-7\gamma^2$$

$$q=\frac{\alpha\beta}{2}+2\beta\gamma+\alpha\gamma=-3\gamma^2-6\gamma^2+2\gamma^2=-7\gamma^2$$

$$p=q, r=-\alpha\beta\gamma=6\gamma^3 \Rightarrow \gamma=\left(\frac{r}{6}\right)^{\frac{1}{3}}$$

$$p=q=-7\left(\frac{r}{6}\right)^{\frac{2}{3}} \Rightarrow \left(\frac{q}{7}\right)^3+\left(\frac{r}{6}\right)^2=0$$

40. D

41. BCD

Sol. (A) $D=0$ therefore range cannot be $[0,\infty)$ because leading coefficient 'a' must also be positive

(B) $x=0,\pi,2\pi \quad y=0,\pi,2\pi \therefore 9$ ordered pairs

(C) $a^2-c^2-b^2=-bc$

$2bc\cos A=bc$

$$A=\frac{\pi}{3}$$

$AM \geq GM$ minimum value is 4

42. AB

Sol. $$a_n=\frac{a_{n-2}\cdot a_{n-1}}{2a_{n-2}-a_{n-1}}$$

$$\Rightarrow a_{n-1}=\frac{2a_{n-2}\cdot a_n}{a_{n-2}+a_n}$$

$\Rightarrow a_{n-2}, a_{n-1}, a_n \rightarrow H.P.$

43. ACD

Sol. Domain of f is $\mathbb{R}$ and since $\sin x$ and $\cos x$ repeats its value

$\therefore f$ is not injective.

Also, $0 \leq \log_2\left(\frac{3\sin x-4\cos x+15}{10}\right) \leq 1$

$\therefore f(x) \in [0,1]$

$f(x)$ is not surjective.

Now $-t^2-2at+a<0 \quad \forall t \in [0,1]$

$\therefore f(0)<0 \ \& \ f(1)<0$

$\Rightarrow a \in (-1,0)$

Now, $g(x)=\dfrac{2}{2-\sin^2 2x}$ $\qquad (g(x))_{\min}=1$

44. AB

Sol. $P=\dfrac{1}{2}\cos x\left(\sin(y+z)+\sin(y-z)\right)\geq \dfrac{1}{2}\cos x\sin(y+z)=\dfrac{1}{2}\cos^2 x$

But $x=\dfrac{\pi}{2}-(y+z)\leq \dfrac{\pi}{2}-2\cdot\dfrac{\pi}{12}=\dfrac{\pi}{3}$

$\therefore P\geq \dfrac{1}{8}$

$P=\dfrac{1}{2}\cos z\left(\sin(x+y)-\sin(x-y)\right)$

Again $P\leq \dfrac{1}{2}\cos^2 z=\dfrac{1+\cos 2z}{4}$

$P\leq \dfrac{2+\sqrt{3}}{8}$

45. ABD

Sol. Let $f(x)=\sum_{r=1}^{n} a_r \sin(rx)$

$\left|f'(0)\right|=\left|\sum_{r=1}^{n} r\,a_r\right|$ *and* $f(0)=0$

$f'(0)=\lim_{x\to 0}\dfrac{f(x)-f(0)}{x}=\lim_{x\to 0}\dfrac{f(x)}{\sin x}\leq 1$

46. ABC

Sol. $(CA)(CT)=(CS)^2=b^2e^2$

$(PN)(PB)=b^2$

$(S_1Q)(S_2R)=a^2$

$(S_1P)(S_2P)=(b+ey_1)(b-ey_1)$

47. BCD

Sol. Equation of tangents $PA \& PB$ is

$y=mx\pm\sqrt{4m^2-9}$

$(2m+1)^2=4m^2-9$

$\Rightarrow 4m=-10\Rightarrow m=-\frac{5}{2}$

$\Rightarrow y+1=-\frac{5}{2}(x-2)$

$\Rightarrow 2y+2=-5x+10$

$\therefore 5x+2y=8$

Equation of AB

$18x+4y=36$

$\Rightarrow 9x+2y=18$

x Coordinate of B

$4x=10, x=\frac{5}{2}$

48. ABCD

SECTION – C

49. 00003.00

Sol. $I=\int_1^3\left(\tan^{-1}\left(\frac{x^2-5x+6}{x^3-6x^2+12x-7}\right)+\tan^{-1}\left(\frac{1}{x^2-4x+4}\right)\right)dx$

$=\int_1^3\left(\tan^{-1}\left(\frac{(x-2)^2-(x-2)}{(x-2)^3+1}\right)+\tan^{-1}\left(\frac{1}{(x-2)^2}\right)\right)dx$

$x-2=t\Rightarrow dx=dt$

$=\int_{-1}^1\left(\tan^{-1}\left(\frac{t^2-t}{1+t^3}\right)+\tan^{-1}\left(\frac{1}{t^2}\right)\right)dt$

$=\int_{-1}^1\left(\tan^{-1}(t^2)-\tan^{-1}(t)+\tan^{-1}\left(\frac{1}{t^2}\right)\right)dt$

$=\frac{\pi}{2}(1+1)=\pi$

$\therefore [I]=3$

50. 00001.00

Sol. $f(y)=\lim_{n\to\infty}\sum_{n=1}^{n}\frac{e^y e^n(e-1)}{(e^n-1)(e^{n+1}-1)}$

$=e^y\lim_{n\to\infty}\sum_{n=1}^{n}\left(\frac{(e^{n+1}-1)-(e^n-1)}{(e^n-1)(e^{n+1}-1)}\right)$

$=e^y\lim_{n\to\infty}\sum_{n=1}^{n}\left(\frac{1}{e^n}-\frac{1}{e^{n+1}-1}\right)$

$$= e^y \lim_{n\to\infty}\left(\left(\frac{1}{e-1}-\frac{1}{e^2-1}\right)+\left(\frac{1}{e^2-1}-\frac{1}{e^3-1}\right)+.....+\left(\frac{1}{e^n-1}-\frac{1}{e^{n+1}-1}\right)\right)$$

$$= \frac{e^y}{e-1}$$

$$\int_0^1 f(x)dx = \frac{1}{e-1}\int_0^1 e^x dx = 1$$

51. 00018.00

Sol. $f(x) = x^3 - 6x^2 + ax - 2a + 17$

$f'(x) = 3x^2 - 12x + a$

$f''(x) = 6x - 12$

$f''(x) = 0 \Rightarrow x = 2$

$f(2) = 1$

$(2,1)$ is the point of inflection on the graph

$y = f(x)$

Straight line $y = 100x - 199$ is also passing through $(2,1)$

$x_1 + x_3 = 4 \; and \; x_3 - x_1 = 6$

$$\int_{x_1}^{x_3} \left(f(x) - 98 + 198 \right) dx$$

$$= \int_{x_1}^{x_3} \left(f(x) - 100x + 199 + 2x - 1 \right) dx$$

$$\int_{x_1}^{x_3} (2x - 1) dx$$

$$= \left(x^2 - x\right)_{x_1}^{x_3} = x_3^2 - x_3 - \left(x_1^2 - x_1\right)$$

$$= (x_3 - x_1)(x_3 + x_1) - (x_3 - x_1)$$

$$= 4 \times 6 - 6 = 18$$

$$\therefore \lambda = 18$$

52. 00000.00

Sol. $S_n = \sum_{k=0}^{n} (-4)^k \; {}^{n+k}C_{2k}$

$S_{n+1} = \sum_{k=0}^{n+1} (-4)^k \; {}^{n+k+1}C_{2k}$

$S_{n-1} = \sum_{k=0}^{n-1} (-4)^k \; {}^{n-1+k}C_{2k}$

Consider ${}^{n+1}C_r + {}^{n-1}C_r - 2\,{}^{n}C_r$ = coefficient of x^r in $\{(1+x)^{n+1} + (1+x)^{n-1} - 2(1+x)^n\}$

= coefficient of x^r in $(1 + x)^{n-1}\{(1 + x)^2 + 1 - 2(1 + x)\}$
= coefficient of x^r in $(1 + x)^{n-1}\{x^2 + 2x + 2 - 2x - 2\}$
= coefficient of x^{r-2} in $(1 + x)^{n-1} = {}^{n-1}C_{r-2}$
Put $n \to n + k$ and $r \to 2k$
$\Rightarrow {}^{n+k+1}C_{2k} + {}^{n+k-1}C_{2k} - 2\,{}^{n+k}C_{2k} = {}^{n+k-1}C_{2(k-1)}$.
Multiplying by $(-4)^k$ and applying summation on both sides

$$\sum_{k=0}^{n+1}(-4)^k\ {}^{n+k+1}C_{2k} + \sum_{k=0}^{n-1}(-4)^k\ {}^{n+k-1}C_{2k} - 2\sum_{k=0}^{n}(-4)^k\ {}^{n+k}C_{2k} = \sum_{k=1}^{n+1}(-4)^k\ {}^{n+k-1}C_{2(k-1)}$$

(Note that the limits of summation are different for each term)

$$\Rightarrow S_{n+1} + S_{n-1} - 2S_n = \sum_{k=1}^{n+1}(-4)^k\ {}^{n+k-1}C_{2(k-1)}$$

Let $k - 1 = t$

$$\sum_{t=0}^{n}(-4)^t(-4)^1\ {}^{n+t}C_{2t} = -4S_n$$

$S_{n+1} + 2S_n + S_{n-1} = 0$
Put $n = 2010$.

53. 00017.00

54. 00002.00

Sol. $\cos^{-1}\dfrac{1-x^2}{1+x^2} + \tan^{-1}\dfrac{2x}{1-x^2} = \pi/3$

Put $x = \tan\theta$, $-\dfrac{\pi}{2} < \theta < \dfrac{\pi}{2}$

$\cos^{-1}(\cos\ 2\theta) + \tan^{-1}(\tan\ 2\theta) = \pi/3$

For $-\pi < 2\theta < -\pi/2$

$-2\theta + 2\theta + \pi = \pi/3$ (Not possible)

For $\dfrac{-\pi}{2} < 2\theta \le 0$

$-2\theta + 2\theta = \pi/3$ (Not possible)

For $0 < 2\theta < \pi/2$

$2\theta + 2\theta = \pi/3$

$\theta = \dfrac{\pi}{12}$

$x = 2 - \sqrt{3}$

For $\dfrac{\pi}{2} < 2\theta < \pi$

$2\theta + 2\theta - \pi = \pi/3 \Rightarrow \theta = \pi/3$

$x = \sqrt{3}$

sum $= 2 - \sqrt{3} + \sqrt{3} = 2$